Experimental Methods in Computer Science

(Metodologias Experimentais em Informática)

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Hypothesis Testing

Hypothesis testing slides are mainly based on chapter 8 of the book "Essentials of Social Statistics for a Diverse Society" Second Edition by Anna Leon-Guerrero, Chava Frankfort-Nachmias , SAGE Publications, Inc, 2010.

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Hypothesis testing steps: **a more pragmatic approach**

Approach already study:

1. State the hypothesis or claim to be tested
2. Select the criteria for a decision (e.g., $\alpha = 0.05$)
3. Compute the test statistic
4. Make a decision

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Hypothesis testing steps: **a more pragmatic approach**

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Hypothesis testing steps: **a more pragmatic approach**

Pragmatic approach:

1. State the hypothesis or claim to be tested
2. Compute the test statistic
3. Obtain p value
4. Make a decision

Hypothesis testing scenario 1 (test for a mean)

Assume you are the database administrator of a big information system and you are unhappy with the execution time of a given SQL package.

From historical data (thousands of previous package executions), you know that the average execution time of the package is **83.54** seconds with a standard deviation of **16.36**.

You change the tuning of the database and run the package several times to check the effect.

Questions:

- **Has the new tuning any effect?**
- **Is the new configuration better?**

That is, is the execution time in the new configuration smaller than in the previous one?

Package exec. time
74
66
88
68
⋮
87
79
78
72
86
85
86

32 times

Avg = 78.15

Example 2 - Step 1: State the hypothesis

(test for a **mean, directional**, known population; normal Z distribution)

- **H_0** – The new configuration has no effect on the execution time of the SQL packaged. → The average execution time is 83.54
- **H_1** – The execution time of the SQL packaged is **smaller** in the new configuration

We are testing whether the null hypothesis H_0 is true

Note that only the alternate hypothesis changed.

Directional or one-tailed tests are hypothesis tests where the alternative hypothesis is stated as greater than (>) or less than (<) the value stated in the null hypothesis

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Example 2 - Step 2: Compute the test statistic

(test for a **mean, directional**, known population; normal Z distribution)

Test statistic:

$$Z_c = \frac{M - \mu}{\sigma / \sqrt{n}} = \frac{78.15 - 83.54}{16.36 / \sqrt{32}} = -1.86$$

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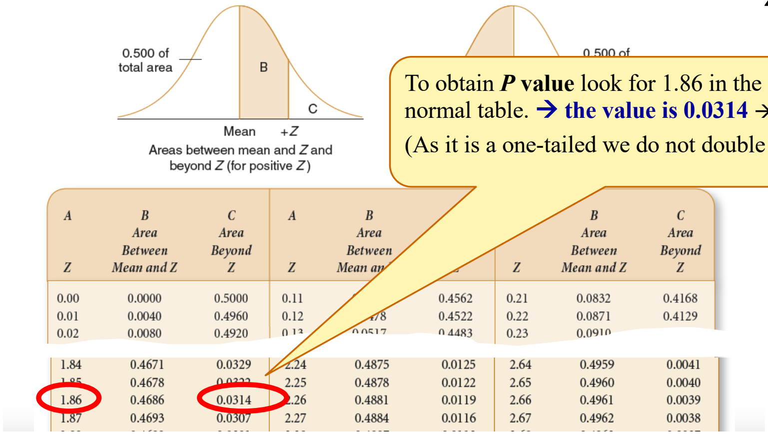
Example 2 - Step 3: Obtain the p value

(test for a **mean, directional**, known population; normal Z distribution)

Standard normal table example

Z = 1.86



A Z	B Area Between Mean and Z	C Area Beyond Z	A Z	B Area Between Mean and Z	C Area Beyond Z	A Z	B Area Between Mean and Z	C Area Beyond Z
0.00	0.0000	0.5000	0.11	[illegible]	0.4562	0.21	0.0832	0.4168
0.01	0.0040	0.4960	0.12	[illegible]	0.4522	0.22	0.0871	0.4129
0.02	0.0080	0.4920	0.13	0.0517	0.4483	0.23	0.0910	[illegible]
1.84	0.4671	0.0329	2.24	0.4875	0.0125	2.64	0.4959	0.0041
1.85	0.4678	0.0322	2.25	0.4878	0.0122	2.65	0.4960	0.0040
1.86	0.4686	0.0314	2.26	0.4881	0.0119	2.66	0.4961	0.0039
1.87	0.4693	0.0307	2.27	0.4884	0.0116	2.67	0.4962	0.0038

To obtain ***P* value** look for 1.86 in the standard normal table. → **the value is 0.0314** → **p = 3.14%**
(As it is a one-tailed we do not double the p)

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Example 2 - Step 4: Make a decision

(test for a **mean, directional**, known population; normal Z distribution)

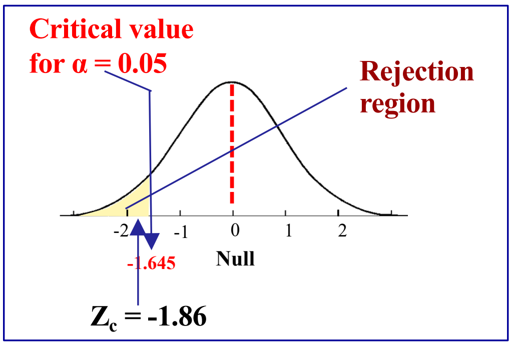
The probability of obtaining $\mathbf{Z_c = -1.86}$ is given by the ***p* value**.

Since **p = 0.0314** this means that the probability of getting an average of 78.15 if H_0 is true is 3.14%

Then I can **reject the H_0** with at least 95% of confidence

The execution time of the SQL packaged is smaller in the new configuration

Classic view



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Types of errors

The conclusion in Step 4 could be wrong, as we are looking at a sample with a limited number n of elements

False positive

		Decision	
		Retain H_0	Reject H_0
Truth in the population	True	**Correct** $1 - \alpha$	**Type I error** α
	False	**Type II error** β	**Correct** $1 - \beta$ **(Power)**

False negative

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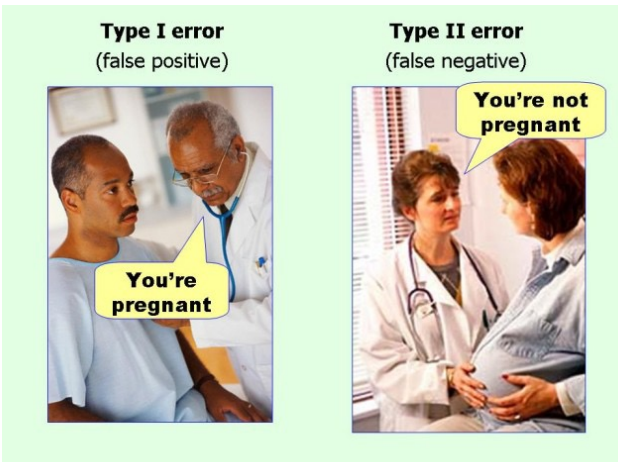
Types of errors



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Some errors are worse than others

The incorrect decision is to retain a false null hypothesis. This is equivalent of doing nothing. We can do more experiments and test again the null hypothesis. Not so bad…

ld be wrong, as we are looking
mber n of elements

False positive

		Decision	
		Retain H_0	Reject H_0
Truth in the population	True	**Correct** $1 - \alpha$	**Type I error** α
	False	**Type II error** β	**Correct** $1 - \beta$ **(Power)**

False negative

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Some errors are worse than others

The incorrect decision is to reject a true null hypothesis. This means rejecting a previous notions of truth that are in fact true (this is equivalent to finding an innocent person guilty).

The … s we are looking
at a … nents

False positive

		…ion	
		Retain H_0	Reject H_0
Truth in the population	True	**Correct** $1 - \alpha$	**Type I error** α
	False	**Type II error** β	**Correct** $1 - \beta$ **(Power)**

False negative

Note that you can directly control the probability of a Type I error by stating an alpha level

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The decision we are looking for

Strong conclusion
This is the decision we are looking for when we test the hypothesis. If we test it, it means we have doubts about such hypothesis.

…ng, as we are looking … elements

False positive

		Decision	
		Retai…	Reject H_0
Truth in the population	True	**Correct** $1 - \alpha$	**Type I error** α
	False	**Type II error** β	**Correct** $1 - \beta$ **(Power)**

False negative

Decision we want

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The example again: <u>**some questions**</u>

Assume you a… information s… execution tim…

What should we do if we cannot have a relatively large number of samples?

From historical data (thousands of previous packag… executions), you know that the average execution time of the package is **83.54** seconds with a standard deviation of **16.36**.

You change the … of the database and run the package several ti…

What should we do if we don't know the standard deviation of the population?

Questions:
- **Has the new tuning any effect?**
- **Is the new configuration better?**

Package exec. time
74
66
88
68
⋮
79
78
72
86
85
86

32 times

In these cases we should use the t Test

Avg = 78.15
Std = 7.936

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T-test

- The **t test** follows a Student's T-distribution (if the null hypothesis is true)
- T-test should be applied when:
  - The **sample size is small** ($n < 30$)
  - The populations' **standard deviation is not known**

(when the number of samples is large, t test and z test give similar results)

Hypothesis testing using T-test (one sample)

- Follows the same steps as for the Z test
- The critical value comes from the **T table** (considering $n - 1$ degrees of freedom)
- The **test statistics** is now the t-test (similar formula)

Mean of the sample

Mean of the population (assumed, often stated as a target)

$$t_c = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

Standard deviation of the sample

Number of elements of the sample

Hypothesis testing steps: pragmatic approach

Steps:

1. State the hypothesis or claim to be tested
2. Compute the test statistic
3. Obtain p value
4. Make a decision

$$t_c = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

Example 3 - Hypothesis testing using T-test (one sample)

- A professor wants to know if their students are proficient in C programming. The professor wants the class to be able to score above 70 (0-100 scale) on the test, but doesn't want to examine all the students.
- The professor selects 6 students at random from the class and give them a C programming test.
- The six students get scores of 62, 92, 75, 68, 83, and 95.
- **Can the professor have 90% confidence that the mean score for the class on the test would be above 70?**

Example 3: **t test** (one sample)

Step 1- State the hypothesis

- **H_0: $\mu \leq 70$**
  In words: the class knows how to program in C with a proficiency not better than 70 in the C programming test
- **H_1: $\mu_1 > 70$**
  The class is better on C programming than the score of 70

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Example 3: **T test** (one sample)

Step 2 - Compute the test statistic

- Average of the sample: 79.17
- Standard deviation of the sample: 13.17

Test statistic:

$$t_c = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{79.17 - 70}{\frac{13.17}{\sqrt{6}}} = 1.71$$

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Example 3: **T test** (one sample)
Step 3 – Obtain p value

- Co
- Lo tai
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- Lo
  - –
  - –

df = 5

In a non-directional two-tailed test, we would divide the **α** value in half

The probability of obtaining **t = 1.71** is given by the ***p* value**. To obtain the ***P* value** look for 1.71 in the **t table**, for **df = 5**

Table entry for *p* and *C* is the critical value *t** with probability *p* lying to its right and probability *C* lying between −*t** and *t**.

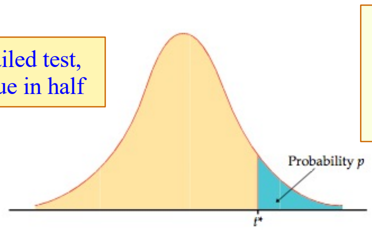

TABLE D

t distribution critical values

	Upper-tail probability *p*											
df	.25	.20	.15	.10	.05	.025	.02	.01	.005	.0025	.001	.0005
1	1.000	1.376	1.963	3.078	6.314	12.71	15.89	31.82	63.66	127.3	318.3	636.6
2	0.816	1.061	1.386	1.886	2.920	4.303	4.849	6.965	9.925	14.09	22.33	31.60
3	0.765	0.978	1.250	1.638	2.353	3.182	3.482	4.541	5.841	7.453	10.21	12.92
4	0.741	0.941	1.190	1.533	2.132	2.776	2.999	3.747	4.604	5.598	7.173	8.610
5	0.727	0.920	[illegible]	1.476	2.015	2.571	2.757	3.365	4.032	4.773	5.893	6.869
6	0.718	0.906	1.134	1.440	1.943	2.447	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	5.959
7	0.711	0.896	1.119	1.415	1.895	2.365	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	5.408
8	0.706	0.889	1.108	1.397	1.860	2.306	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	5.041
9	0.703	0.883	1.100	1.383	1.833	2.262	2.398	2.821	3.250	3.690	4.297	4.781
10	0.700	0.879	1.093	1.372	1.812	2.228	2.359	2.764	3.169	3.581	4.144	4.587
11	0.697	0.876	1.088	1.363	1.796	2.201	2.328	2.718	3.106	3.497	4.025	4.437
12	0.695	0.873	1.083	1.356	1.782	2.179	2.303	2.681	3.055	3.428	3.930	4.318
13	0.694	0.870	1.079	1.350	1.771	2.160	2.282	2.650	3.012	3.372	3.852	4.221
14	0.692	0.868	1.076	1.345	1.761	2.145	2.264	2.624	2.977	3.326	3.787	4.140

→ **t score = 1.71**

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Example 3: **T test** (one sample)
Step 4 - Make a decision

The probability of obtaining **t = 1.71** is given by the ***p* value**. To obtain the ***P* value** look for 1.71 in the **t table**, for **df = 5**

→ the P value is between 5% and 10% (P = 7.4%)

As *p* < 10% → **Reject the null hypothesis (reach significance)**

Means that the probability of getting an average score of 79.17 if $\mathbf{H_0}$ **is true** is 7.4%

Conclusion: The class is better on C programming than the score of 70

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Measuring the size of an effect

- A decision to reject the null hypothesis means that **an effect is significant**. Hypothesis testing does not inform on how big the effect is.
- **Effect size** is a statistical measure of the size of an effect in a population. It particularly makes sense when the null hypothesis is rejected.
- **Cohen's *d*** measures the number of standard deviations an effect shifted above or below the population mean stated by the null hypothesis

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Cohen's d measure formula

Mean of the sample

Mean of the population

Standard deviation of the population

$$\textbf{Cohen's d} = \frac{M - \mu}{\sigma}$$

Cohen's effect size conventions are often used to interpreter the effect size

If values of *d* are negative, the effect shifted below the population mean

Description of Effect	Effect Size (*d*)
Small	$\lvert d \rvert < 0.2$
Medium	$0.2 < \lvert d \rvert < 0.8$
Large	$\lvert d \rvert > 0.8$

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The example again: *Cohen's d*

Assume you are the database administrator of a big information system and you are unhappy with the execution time of a given SQL package.

From historical data (thousands of previous package executions), you know that the average execution time of the package is **83.54** seconds with a standard deviation of **16.36**.

You change the tuning of the database and run the package several times to check the effect.

Package exec. time
74
66
88
68
⋮
79
78
72
86
85
86

32 times

Avg = 78.15

$$\textbf{Cohen's d} = \frac{M-\mu}{\sigma} = \frac{78.15-83.54}{16.36} = -0.33$$

The observed effect shifted 0.33 standard deviations below the mean

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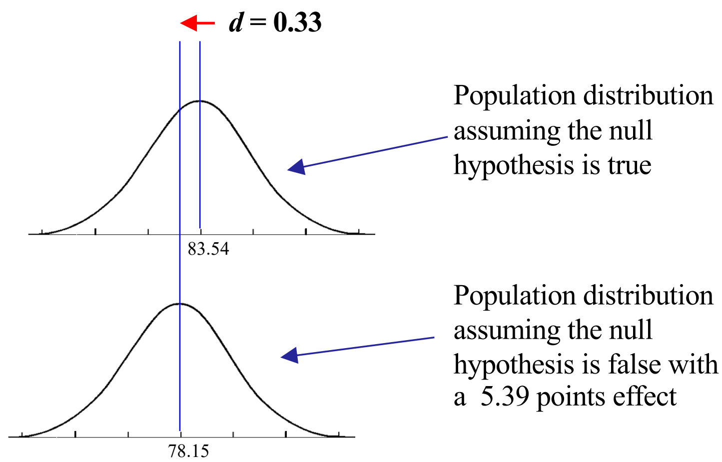
The example again: *Cohen's d*



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T-test

- The **T test** follows a Student's T-distribution (if the null hypothesis is true)
- Two types:
  - **One-sample T-tests** → used to compare a sample mean with the known population mean
  - **Two-sample T-tests** → used to compare two samples.
- T-test should be applied when:
  - The **sample size is small** (n < 30)
  - The populations' **standard deviation is not known**

Independent samples: unrelated separate groups

(when the number of samples is large, t test and z test give similar results)

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Hypothesis testing using T-test (two samples)

- Follows the same steps as for the Z test
- The critical value comes from the **T table** (the degrees of freedom is the smaller n_1 -1 and n_2 -1)
- The **test statistics** is now the **two sample T-test**

Means of the two samples

Hypothesized difference between the population means (0 if testing for equal means)

$$t_c = \frac{\bar{x}_1 - \bar{x}_2 - \Delta}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Standard deviation of the two samples

Number of elements of the two samples

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Example 4 - Hypothesis testing using T-test (two independent samples)

Assume you are the database administrator of a big information system. The database has just been installed and you are trying two tuning configurations: Conf. **A** and Conf. **B**.

You use a given SQL package to test the execution time for each configuration.

After running several times the SQL package in both configurations you want to take a decision.

Important: we consider that the measurement samples obtained with each configuration are independent.

Question: **what is the best configuration?**

Conf. A exec. time	Conf. B exec. time
74	69
66	71
88	80
68	88
79	64
68	65
87	74
79	76
78	89
72	68
86	67
85	72
86	

$\mu_1 = 78.15$ $\mu_2 = 73.58$
$s_1 = 7.94$ $s_2 = 8.33$
n = 13 **n** = 12

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Example 4: **t test** (two indep. samples)
Step 1- State the hypothesis

- **H_0: $\mu_1 = \mu_2$**
  In words: configuration A and B are equivalent concerning the execution time of the SQL package
- **H_1: $\mu_1 > \mu_2$**
  Configuration B is faster than configuration A (i.e., the execution time of the SQL package is higher in configuration A)

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Example 4: **T test** (two indep. samples)
Step 2 - Compute the test statistic

Sample	Configuration	n	*x*	s
1	A	13	78.15	7.94
2	B	12	73.53	8.33

Test statistic:

$$t_c = \frac{\bar{x}_1 - \bar{x}_2 - \Delta}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{78.15 - 73.58 - 0}{\sqrt{\frac{7.94^2}{13} + \frac{8.33^2}{12}}} = 1.402$$

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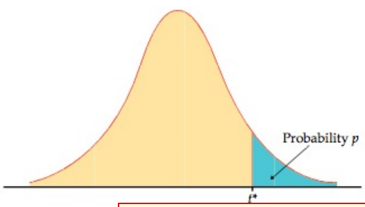
Example 4: **t test** (two indep. samples)
Step 3 – Obtain the p value

As the size of the samples are n = 13 and n = 12, the degree of freedom is the smaller n -1 → df = 11

T score = 1.402
df = 11

Table entry for p and C is the critical value t^* with probability p lying to its right and probability C lying between $-t^*$ and t^*.



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2	0.816	1.061	1.386	1.886	2.920	4.303	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
3	0.765	0.978	1.250	1.638	2.353	3.182	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
4	0.741	0.941	1.190	1.533	2.132	2.776	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
5	0.727	0.920	1.156	1.476	2.015	2.571	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
6	0.718	0.906	1.134	1.440	1.943	2.447	2.612	3.143	3.707	4.317	5.208	5.959
7	0.711	0.896	1.119	1.415	1.895	2.365	2.517	2.998	3.499	4.029	4.785	5.408
8	0.706	0.889	1.108	1.397	1.860	2.306	2.449	2.896	3.355	3.833	4.501	5.041
9	0.703	0.883	1.100	1.383	1.833	2.262	2.398	2.821	3.250	3.690	4.297	4.781
10	0.700	0.879	1.093	1.372	1.812	2.228	2.359	2.764	3.169	3.581	4.144	4.587
11	[illegible]	[illegible]	[illegible]	1.363	1.796	2.201	2.328	2.718	3.106	3.497	4.025	4.437
12	0.695	0.873	1.083	1.356	1.782	2.179	2.303	2.681	3.055	3.428	3.930	4.318
13	0.694	0.870	1.079	1.350	1.771	2.160	2.282	2.650	3.012	3.372	3.852	4.221
14	0.692	0.868	1.076	1.345	1.761	2.145	2.264	2.624	2.977	3.326	3.787	4.140
15	0.691	0.866	1.074	1.341	1.753	2.131	2.249	2.602	2.947	3.286	3.733	4.073

To obtain ***P* value** look for 1.402 in the **T table**, for **df = 11.**

The **p value is between 0.05 and 0.1**

→ a more accurate P value from an online calculator is 0.0942 (P = 9.42%)

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Example 4: **T test** (two indep. samples) Step 4 - Make a decision

p = **0.0942 (P = 9.42%)**

Means that the probability of getting an average score of 73.58 if H_0 is true is 9.42%

As *p* > 5% → **Retain the null hypothesis (fail reaching significance)**

We could not proved that configuration B is faster than A with 95% confidence

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Hypothesis testing steps: summary of test statistics (so far…)

Pragmatic approach:

1. State the hypothesis or claim to be tested
2. Compute the test statistic
3. Obtain p value
4. Make a decision

When the sample size is large (n > 30) and the σ of the population is known

$$Z_c = \frac{M - \mu}{\sigma / \sqrt{n}}$$

One sample Z test – to compare a sample mean with the population mean

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Hypothesis testing steps: summary of test statistics (so far…)

Pragmatic approach:

1. State the hypothesis or claim to be tested
2. Compute the test statistic
3. Obtain p value
4. Make a decision

When the size of the samples is large ($n \geq 30$)

$$Z_c = \frac{\bar{x}_1 - \bar{x}_1 - \Delta}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Two samples Z test – to compare the means of two independent large samples

Hypothesis testing steps: summary of test statistics (so far…)

Pragmatic approach:

1. State the hypothesis or claim to be tested
2. Compute the test statistic
3. Obtain p value
4. Make a decision

When the sample size is small ($n < 30$) and

$$t_c = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

One sample T test – to compare a sample mean with the population mean

Hypothesis testing steps: summary of test statistics (so far…)

Pragmatic approach:

1. State the hypothesis or claim to be tested
2. Compute the test statistic
3. Obtain p value
4. Make a decision

When the size of the samples is small (n < 30)

$$t_c = \frac{\bar{x}_1 - \bar{x}_2 - \Delta}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Two samples T test – to compare the means of two independent samples

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Hypothesis testing steps: obtaining p value

Pragmatic approach:

1. State the hypothesis o
2. Compute the test s
3. Obtain p value
4. Make a decision

The test statistic is converted into a conditional probability, the **p value**. It can be obtained using the t tables or using p value calculation sites/programs.

The p value answers the question "**If the null hypothesis is true, what is the probability of observing the measured data?**"

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Hypothesis testing steps: p value conventions for a decision

Small ***p*** values provide evidence against the null hypothesis, as it means that the observed data are unlikely when the null hypothesis is true.

Conventions:

- $p \geq 0.10$ → **the observed difference is "not significant"**
- $0.05 \leq p < 0.10$ → **the observed difference is "marginally significant"**
- $0.01 \leq p < 0.05$ → **the observed difference is "significant"**
- $p < 0.01$ → **the observed difference is "highly significant"**

P

1.

2.

3. Obtain

4. Make a decision

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Hypothesis Testing

Test for a proportion

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